



Servlet II

1. Handling request parameters
2. Config and context parameters
3. Request dispatching
4. Request redirection

- Any number of request parameters can be sent by the client while making the request.
- From the request object we can read the parameter
- If parameter has only one value associated, we can use the method `getParameter("parameter_name")` which returns `String`
- If parameter has only multiple values associated, we can use the method `getParameterValues("parameter_name")` which returns array of strings
- If `parameter_name` is not found in the request, it will return `null`.

- Both ServletContext and ServletConfig are basically the configuration objects
- They have some difference in terms of scope and availability.
- What is ServletConfig:
 - For each Servlet under execution, a ServletConfig object is created by the servlet container
 - It is used by the programmer to read the servlet specific data declared in the web.xml or in annotation @WebInitParam
 - The getServletConfig() method of the GenericServlet class returns an object of the ServletConfig.
 - The ServletConfig object is specific for a Servlet.

- What is ServletContext?
 - The ServletContext object is common to all servlets and other components
 - ServletContext parameters are defined in web.xml inside the `<context-param />` tag
 - The ServletContext object comes into existence when the container is started and is destroyed when the container stops its execution.
- Both ServletConfig and ServletContext provide the method `getInitParameter("parameter_name")` which returns the configured value in the form of string.
- Difference between ServletConfig and ServletContext

- It is a technique that allows for 'including' content in a request/response or 'forwarding' a request/response to a resource.
- This technique helps in dividing the task on the server side into different web components.
- Request dispatching supports 2 methods:
- **forward method**
 - Forwards the request from one servlet to another resource.
 - This method will close the output stream after it has been invoked.
- **include method**
 - It includes the content of the resource(such as servlet, JSP, HTML file) in the response.
 - This method leaves the output stream open
- Request dispatching gets executed withing one request-response cycle

- Request redirection is a technique which redirects the user from one URL to another.
- The `sendRedirect()` method
 - This method of `HttpServletResponse` interface can be used to redirect the response to another resource i.e. it may be a Servlet, JSP or HTML file.
- Client receives a redirection url in the response which will be requested once again by the browser
- Comparison between forward and `sendRedirect`